CSC1024 Programming Principles
Coding Exercise 5

1. Improve the following Python program (Sample Answer for CE04 Q5) as shown in Listing 1 by:
   - Randomising the `secret_num`. The random number generated must be within the range of 1 to 200.
   - Add an iterative function using the while loop to ask the user for a decision to continue playing the game.

```
# Question 5
secret_num = 50
guess_num = int(input('Guess my secret number: '))
count = 1

while guess_num != secret_num:
    if guess_num < secret_num:
        print( 'Too Low')
    else:
        print('Too High')
    count = count + 1
    guess_num = int(input('Have another guess: '))

print('Well done, you took ' + str(count) + ' attempts.')
```

Listing 1

2. Write a Python program using the while loop to generate a multiplication table that will have the following outputs.

```
Which multiplication table would you like to print? 8
How high would you like it to go? 10

Here is your multiplication table:
8 times 1  =   8
8 times 2  =  16
8 times 3  =  24
8 times 4  =  32
8 times 5  =  40
8 times 6  =  48
8 times 7  =  56
8 times 8  =  64
8 times 9  =  72
8 times 10 =  80
```

3. Write a Python program using the while loop to print the following pattern.

```
10101010101
10101010101
10101010101
10101010101
10101010101
```

Do not use the while loop to print the pattern (`'10101010101'`) 5 times. The program should print the pattern character-by-character.

4. Write a temperature conversion program between degree Celsius and degree Fahrenheit, according to user selection. The following is the formula:

```
Celsius = (Fahrenheit – 32) * 5/9
Fahrenheit = (Celsius * 9/5) + 32
```

- The temperature conversion program is expected to convert a range of temperature measurements from a minimum value to a maximum value.
- The program will only execute if the minimum temperature is smaller than or equal to the maximum temperature. Else, an error message should be prompted to the user and the program restart.
- The program will only execute when the menu selection is valid. Else, an error message should be prompted to the user and the program restart.
- Write an iteration check if the user would like to run the program again.

Screen output examples:

```
Temperature Conversion Programme.
[1] Convert Celsius to Fahrenheit.
[2] Convert Fahrenheit to Celsius.
Enter your selection, 1 or 2: 1
Celsius (C) to Fahrenheit (F) Conversion
Enter temperature in integer values only.
Enter minimum temperature: 20
Enter maximum temperature: 25
20C = 68.0F
21C = 69.8F
22C = 71.6F
23C = 73.4F
24C = 75.2F
25C = 77.0F
Conversion Done!
Do you want to run the program again? [Y/N]: Y

Temperature Conversion Programme.
[1] Convert Celsius to Fahrenheit.
[2] Convert Fahrenheit to Celsius.
Enter your selection, 1 or 2: 3
 ERROR: Invalid Selection!
Do you want to run the program again? [Y/N]: Y

Temperature Conversion Programme.
[1] Convert Celsius to Fahrenheit.
[2] Convert Fahrenheit to Celsius.
Enter your selection, 1 or 2: 2
Fahrenheit (F) to Celsius (C) Conversion
Enter temperature in integer values only.
Enter minimum temperature: 50
Enter maximum temperature: 30
ERROR: Invalid Input!
Do you want to run the program again? [Y/N]: Y
```